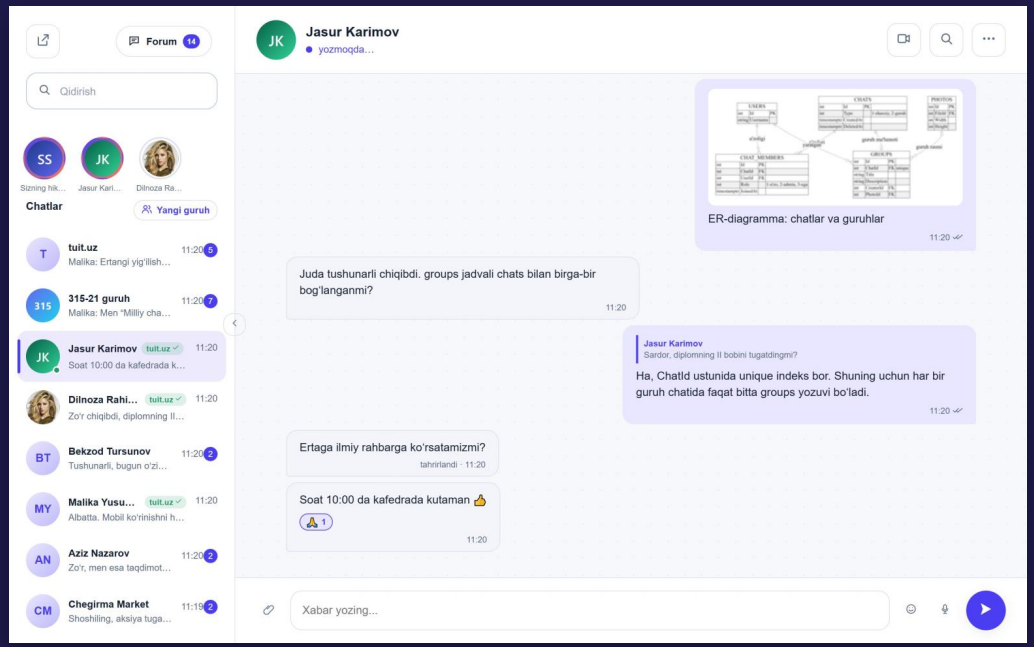
O‘ZBEKISTON RESPUBLIKASI RAQAMLI TEXNOLOGIYALAR VAZIRLIGI
MUHAMMAD AL-XORAZMIY NOMIDAGI TOSHKENT AXBOROT TEXNOLOGIYALARI UNIVERSITETI
“Axborot texnologiyalarining dasturiy ta’minoti” kafedrasi

BITIRUV MALAKAVIY ISHI

Real vaqtda xabar almashish imkonini beruvchi “Milliy” xabarchi ishlab chiqish

Telegramga o‘xshash milliy veb-messenjer



Bajardi: **S.O. Sohinazarov**

Ilmiy rahbar: **dots. Y.Sh. Yuldoshev**

Toshkent – 2026

Mavzuning dolzarbligi

“Raqamli O‘zbekiston – 2030” strategiyasi

PF-6079-son Farmon: mahalliy dasturiy mahsulotlar va axborot xavfsizligi ustuvor vazifa

Ma’lumotlarni lokalizatsiya qilish talabi

“Shaxsga doir ma’lumotlar to‘g‘risida”gi Qonun: fuqarolar ma’lumotlari respublika hududida saqlanishi kerak

Xorijiy messenjerlarga bog‘liqlik

Telegram, WhatsApp, Signal, WeChat ma’lumotlarni chet elda saqlaydi; mahalliy Humo esa pullik va faqat davlat xodimlari uchun

Yechim — “Milliy” xabarchi

- Ma’lumotlar O‘zbekistondagi serverda saqlanadi
- Hamma uchun ochiq va bepul
- O‘z kodi, mahalliy nazoratda
- Tashkilotlar uchun ichki aloqa vositasi

Loyihaning maqsadi va vazifalari

Maqsad: foydalanuvchi ma’lumotlarini o‘z serverida saqlaydigan, shaxsiy va guruh yozishmalarini real vaqtda ta’minlaydigan “Milliy” xabarchi veb-messenjerini loyihalash va ishlab chiqish.

01

Mavjud messenjerlar va real vaqt texnologiyalarini tahlil qilish

02

Tizimga funksional va nofunksional talablarni aniqlash

03

Arxitektura, ma’lumotlar bazasi va UML modellarini loyihalash

04

Xavfsiz autentifikatsiya va E2E shifrlangan maxfiy chatlar

05

Server, klient, lotin↔kirill va tashkilot rejimini ishlab chiqish

06

Bulutga joylashtirish, sinash va HFX talablarini ishlab chiqish

Mavjud messenjerlar tahlili

Mezon	Telegram	WhatsApp	Signal	WeChat	Humo	“Milliy” xabarchi
Egasi (davlat)	BAA	Meta (AQSh)	AQSh	Tencent (Xitoy)	UNICON-SOFT (O‘zb.)	**Mahalliy**
Kimlar uchun, narxi	Hamma, bepul	Hamma, bepul	Hamma, bepul	Hamma, bepul	Davlat xodimlari, pullik	**Hamma, bepul**
Ro‘yxatdan o‘tish	Telefon raqami	Telefon raqami	Telefon raqami	Telefon raqami	—	**E-pochta kodi, Google**
To‘liq shifrlash	Faqat maxfiy chatlarda	Hamma chatlarda	Hamma chatlarda	Yo‘q	E’lon qilingan	**Maxfiy chatlarda**
Ma’lumotlar joyi	Chet elda	Chet elda	Chet elda	Xitoyda	O‘zbekistonda	**O‘zbekistonda ham mumkin**
Lotin ↔ kirill	Yo‘q	Yo‘q	Yo‘q	Yo‘q	—	**Bor (qidiruv ham)**
Tashkilot tasdig‘i	Yo‘q	Yo‘q	Yo‘q	Yo‘q	—	**Ish e-pochtasi**

Xorijiy messenjerlar ma’lumotlarni chet elda saqlaydi, Humo esa pullik va faqat davlat xodimlari uchun. Qulaylik bo‘yicha namuna sifatida Telegram tanlandi.

Nega yana bitta messenjer kerak?

Sun’iy intellekt yordamchisi — ishlab chiqilmoqda. Yozishmaning qisqacha mazmuni, javob takliflari, ovozli xabarni matnga aylantirish, o‘zbek–rus–ingliz tarjima, imloni tuzatish, spam va haqoratni aniqlash.

Funksional ustunliklar

- Lotin va kirill: har kim o‘z yozuvida o‘qiydi, qidiruv ikkalasida
- Tashkilot rejimi: ish e-pochtasi orqali yopiq guruhga
- Maxfiy chatlar: server ham o‘qiy olmaydi
- Forum: ochiq muhokama, obunalar, top lenta
- Parolsiz kirish: SMS va telefon raqami shart emas
- Boshqa tizimlarga qo‘shiladi: Elektron kooperatsiya portali

Nofunksional ustunliklar

- O‘z kodi: Telegram nusxasi emas, noldan yozilgan
- Hamma uchun ochiq va bepul
- Ma’lumotlar tashkilotning o‘z serverida
- O‘rnatish shart emas: brauzerda, telefonda ham
- Sekin internetda ham ishlaydi
- Xavfsiz: sessiyalar nazorati, antivirus, xeshlar
- Kengaytirish oson: bir nechta server, Docker
- 700 dan ortiq avtomatik test

Real vaqtda aloqa texnologiyalari

Texnologiya	Yo‘nalish	Kechikish	Brauzerlar
Short polling	Mijoz → server	Yuqori	Barchasi
Long polling	Mijoz → server	O‘rtacha	Barchasi
Server-Sent Events	Server → mijoz	Past	Zamonaviy
WebSocket	Ikki tomonlama	Juda past	Zamonaviy
SignalR	**Ikki tomonlama**	**Juda past**	**Barchasi (fallback)**

Nega SignalR?

WebSocket → SSE → Long polling

transport avtomatik tanlanadi

- Hab (Hub) abstraksiyasi: server va mijoz bir-birining metodlarini chaqiradi
- Guruhlar, JWT autentifikatsiyasi va qayta ulanish tayyor holda
- .NET tarkibiy qismi; brauzer uchun rasmiy @microsoft/signalr

Tizimga qo‘yilgan talablar

19 **funksional talab**

- Kirish: e-pochta kodi yoki Google
- Profil va faol sessiyalar
- Shaxsiy va guruh chatlari
- Javob, tahrirlash, o‘chirish, qidiruv
- Rasm, video va fayllar
- “Yozmoqda”, o‘qildi, onlayn holat
- Reaksiyalar, bloklash, 24 soatlik lavhalar
- Forum: postlar, obunalar, top lenta
- Desktop bildirishnomalar
- Yorug‘ / qorong‘i mavzu, mobil ko‘rinish
- Lotin↔kirill, tashkilot, maxfiy chat

4 **nofunksional talab**

- Unumdorlik: xabar < 1 soniyada
- Xavfsizlik: HTTPS, xeshlar, chastota cheklovi
- Ishonchlilik va kengaytiriluvchanlik
- Ko‘chiriluvchanlik: Docker

Tizim aktorlari

Mehmon
kirish, ro‘yxatdan o‘tish

Foydalanuvchi
chatlar, forum, lavhalar, reaksiyalar

Guruh egasi
a’zolar va rollarni boshqarish

Texnologiyalar steki

Server qismi

- .NET muhiti
- SignalR — real vaqt
- Entity Framework — baza bilan ishlash
- PostgreSQL — ma’lumotlar bazasi
- Redis — onlayn holat

Klient qismi

- Angular
- TypeScript
- SignalR klienti
- Google orqali kirish
- O‘z dizayn tizimi
- Web Crypto API — maxfiy chatlar

Infratuzilma

- Docker — konteynerlash
- Render bulut xizmati
- Pochta — kirish kodlari
- ClamAV — antivirus
- Rasm va video ishlovi
- Git va GitHub

✓ Barcha texnologiyalar ochiq kodli va bepul — litsenziya xarajati yo‘q.

Tizim arxitekturasi

Clean Architecture — server qatlamlari

bog‘liqlik ichkariga

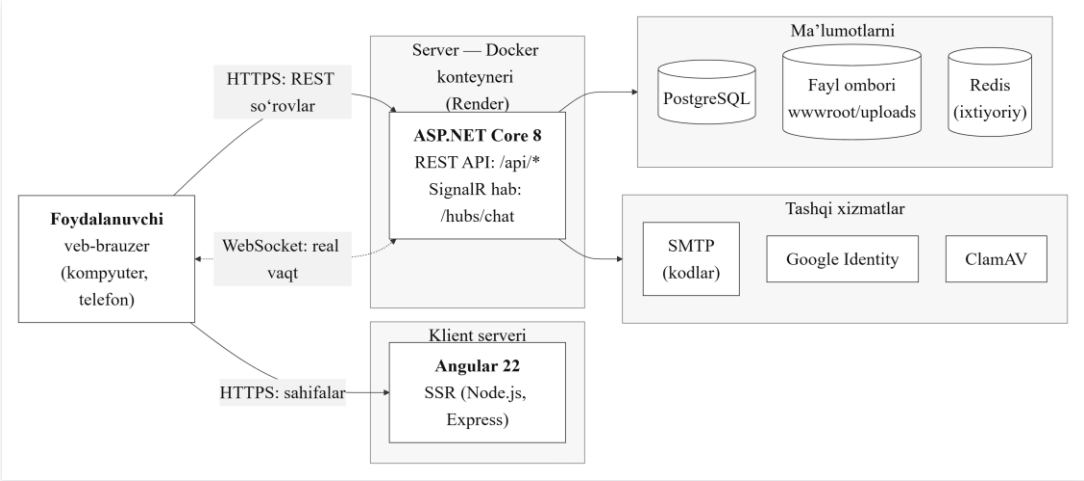
API
Controllerlar, ChatHub (SignalR), middleware

Infrastructure
EF Core, JWT, PBKDF2 / HMAC, SMTP, ClamAV, Redis

Application
Servislar, DTO, FluentValidation, portlar

Domain
User, Chat, Message, Group, Story …

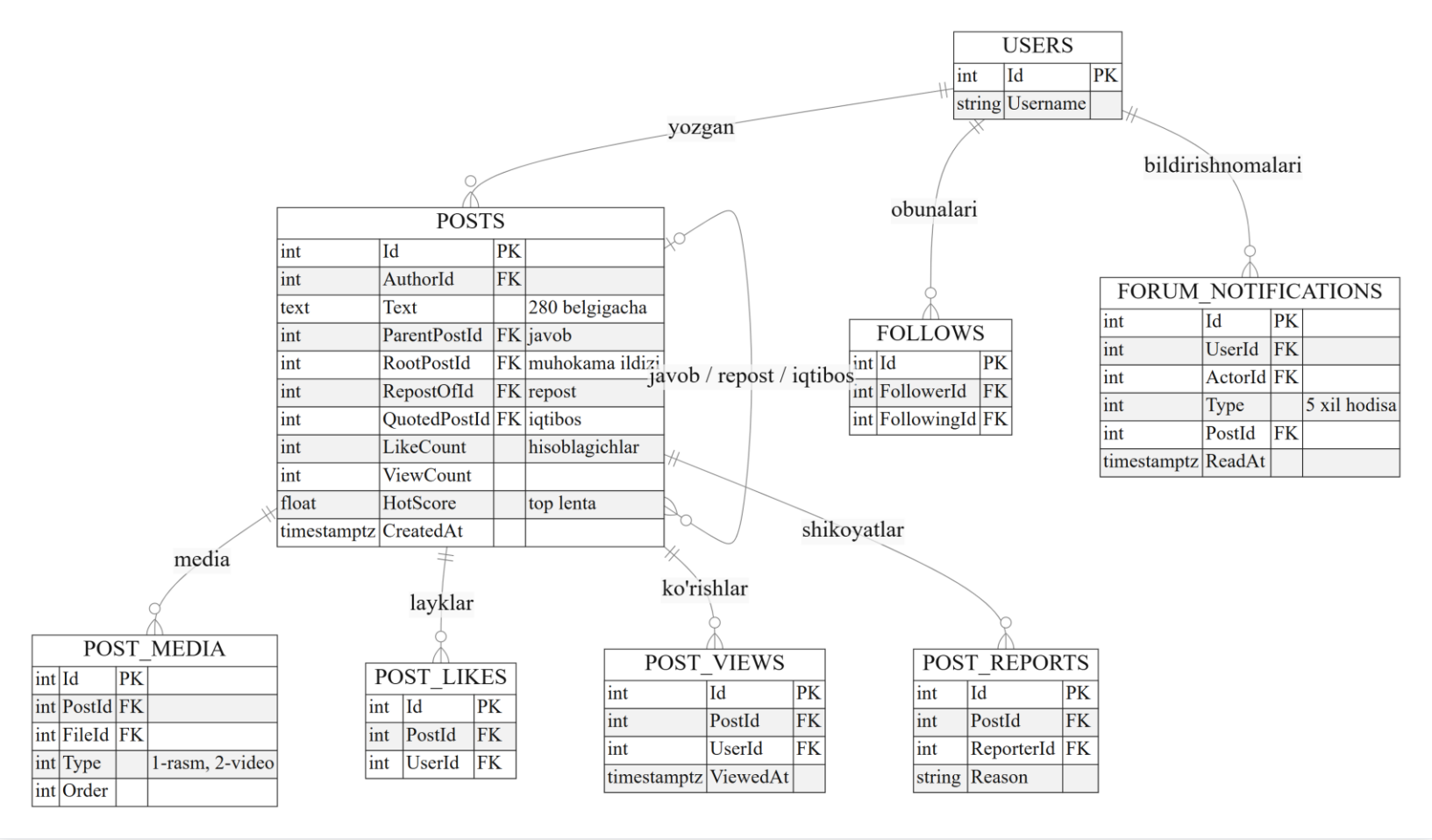
Joylashtirish diagrammasi



Klient (Angular) tuzilmasi

- features/* — har bir bo‘lim alohida: auth, chat, groups, stories
- data-access — HTTP va SignalR servislari
- signals asosida holat, SSR bilan tez ochilish

Ma’lumotlar bazasini loyihalash



11

sxema

identity, messaging, forum, organizations …

43

jadval

kelajakdagi kanallar va botlar uchun ham

Keyset

kursorli sahifalash

beforeId + limit (1–100), totalCount siz

Xavfsizlik va autentifikatsiya

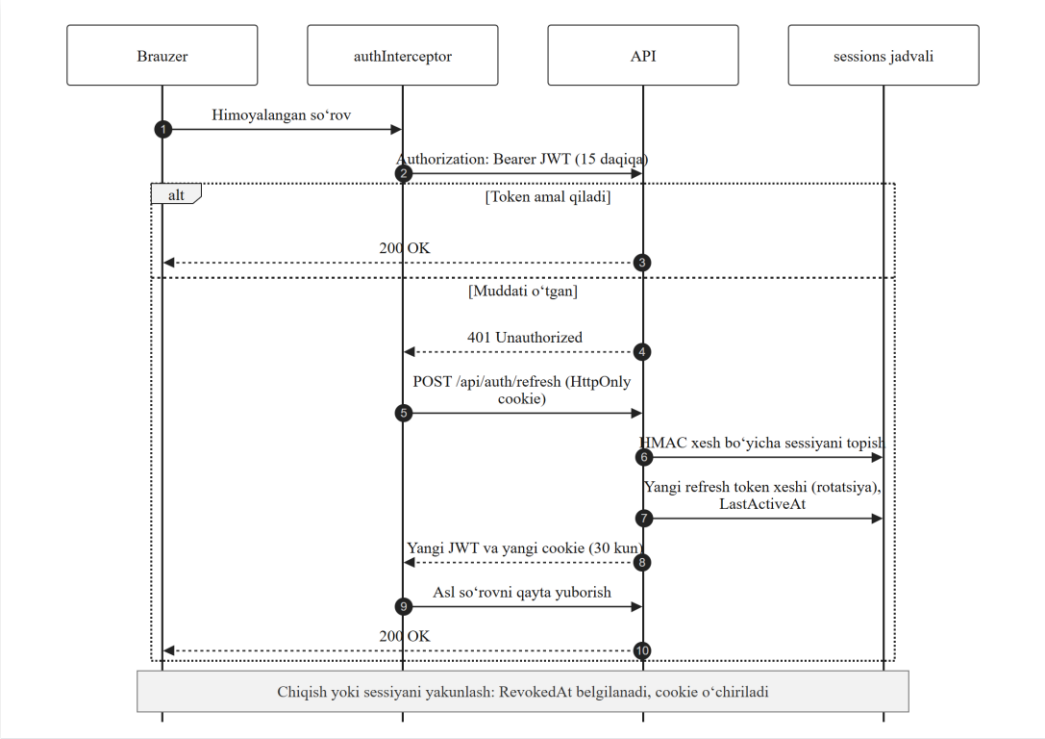
1. **Elektron pochtaga 6 xonali kod**
   Bazada faqat PBKDF2 xeshi saqlanadi, parol umuman yo‘q

2. **Yoki Google hisobi orqali**
   Google ID tokeni serverda tekshiriladi

3. **JWT access token — 15 daqiqa**
   Har bir API so‘rovi va SignalR ulanishida tekshiriladi

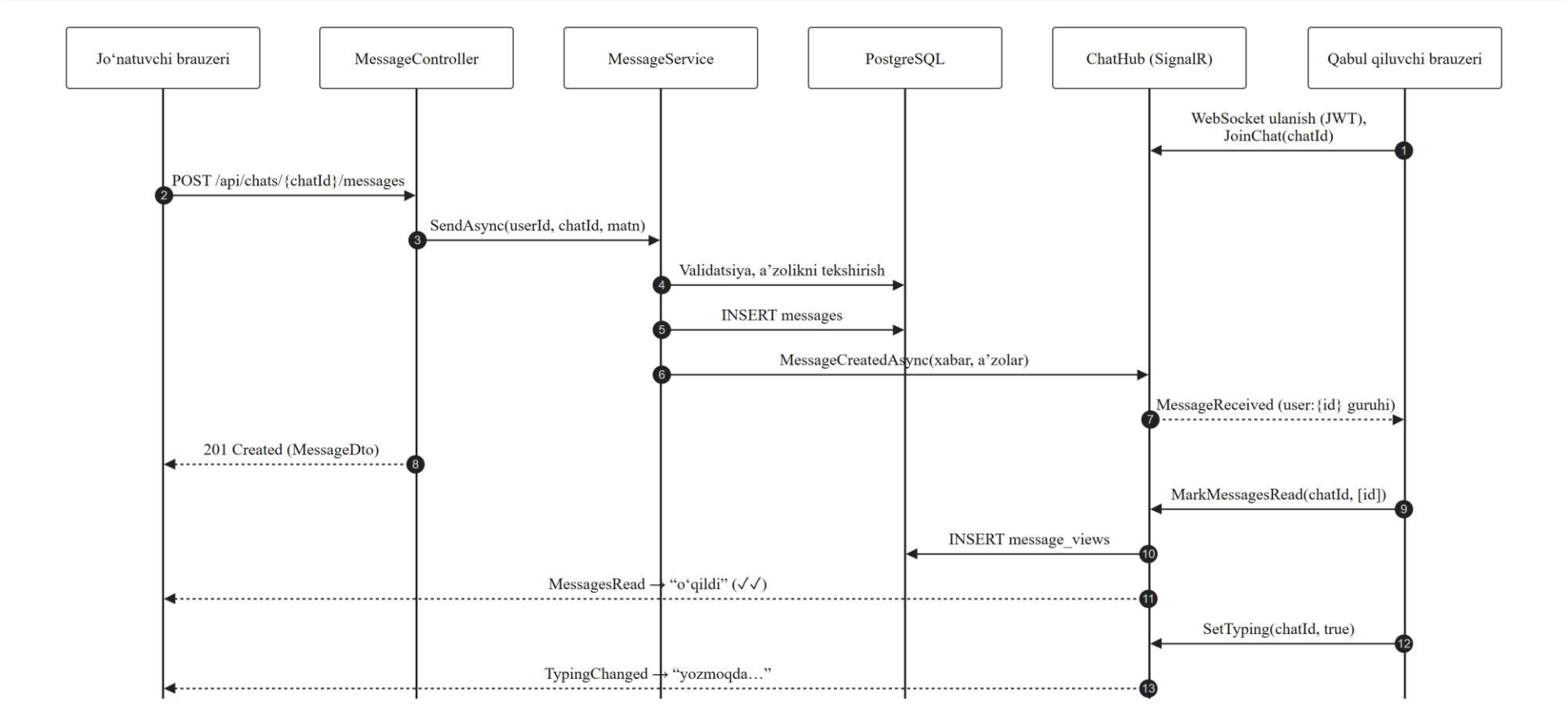
4. **Refresh token — 30 kun, rotatsiya**
   HttpOnly cookie’da, bazada HMAC xeshi; har safar almashadi

5. **Qurilmalar bo‘yicha sessiyalar**
   Bitta yoki boshqa barcha sessiyalarni yakunlash mumkin



- Har so‘rovda chat a’zoligi tekshiriladi
- Fayllar ClamAV antivirusidan o‘tadi

Real vaqtda xabar yuborish jarayoni



1 REST: xabar tekshirilib, bazaga yoziladi

2 ChatHub: chat a’zolariga darhol yuboriladi

3 O‘qildi ✓✓ va “yozmoqda…” belgilari

“Milliy” xabarchining o‘ziga xos imkoniyatlari

Lotin ↔ Kirill

- Har kim xabarni o‘zi tanlagan yozuvda ko‘radi
- Xabar bazada asl holida qoladi
- Qidiruv yozuvga bog‘liq emas

Tashkilot rejimi

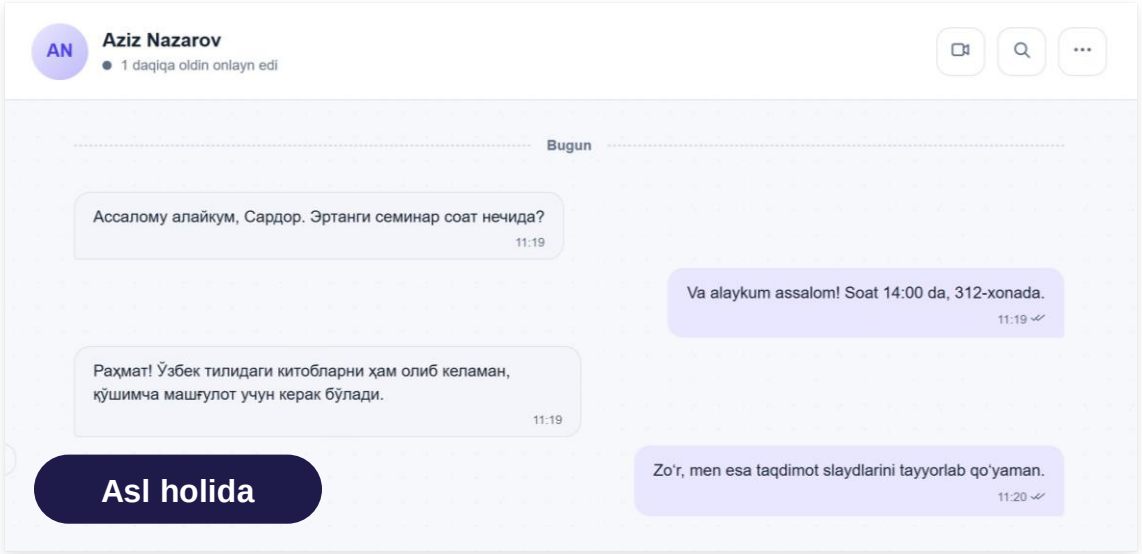
- Tashkilot = e-pochta domeni
- “tuit.uz ✓” belgisi, yopiq domen guruhi
- Gmail va boshqalar faqat taklif havolasi orqali

Maxfiy chat (E2E)

- Server xabar mazmunini ko‘rmaydi
- X25519 + AES-256-GCM, oldinga maxfiylik
- Kalit izi, taymer, shifrlangan fayllar

✓ Birinchi ikki imkoniyat ko‘rib chiqilgan xorijiy messenjerlarning birortasida yo‘q — ular aynan o‘zbek foydalanuvchisi uchun.

Lotin ↔ Kirill: har kim o‘z yozuvida o‘qiydi





- Xabar bazada asl holida saqlanadi, o‘girish faqat ko‘rsatishda
- Kontekst qoidalari: е → ye/e, ц → ts/s, ў → o‘, tutuq belgisi
- URL, @username va kod o‘girilmaydi
- Server (C#) va klient (TS) — bitta 80 juftlikli test to‘plami

```
PS> psql -d NationalChat -c "SELECT Id, TextContent, SearchText FROM messaging.messages WHERE ChatId = 2;"
 Id |             TextContent              |              SearchText
----+--------------------------------------+--------------------------------------
  5 | Ассалому алайкум, Сардор. Эртанги семи | assalomu alaykum, sardor. ertangi semi
  6 | Va alaykum assalom! Soat 14:00 da, 312 | va alaykum assalom! soat 14:00 da, 312
  7 | Раҳмат! Ўзбек тилидаги китобларни ҳам  | rahmat! o'zbek tilidagi kitoblarni ham
  8 | Zo‘r, men esa taqdimot slaydlarini tay | zo'r, men esa taqdimot slaydlarini tay
(4 rows)
```

SearchText — yozuvdan qat’i nazar qidiruv kaliti

Tashkilot rejimi: e-pochta domeni orqali tasdiq

Tasdiqlangan pochta bilan kirish

ali@tuit.uz → tashkilot “tuit.uz”

Birinchi foydalanuvchi — admin

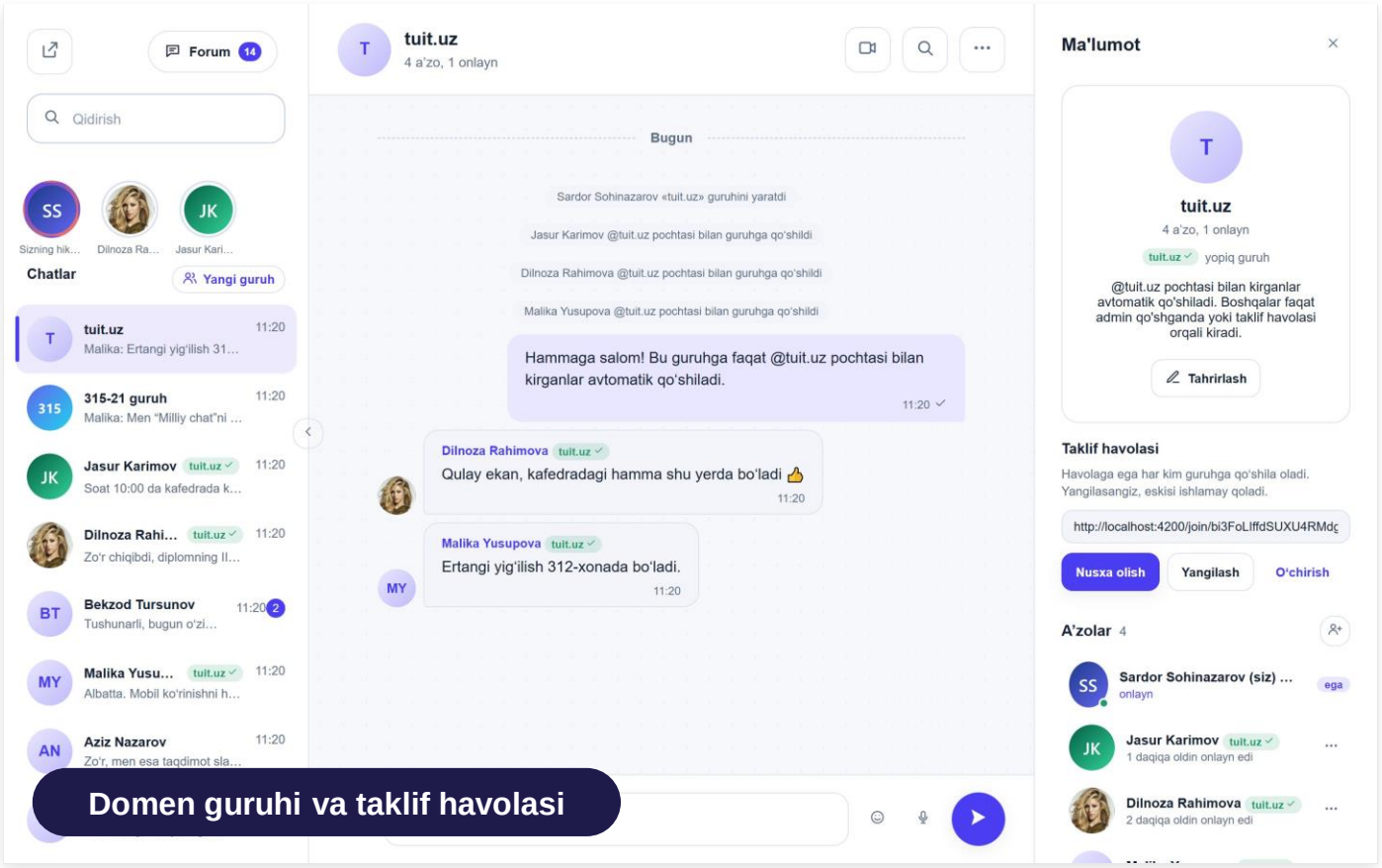
Domenning yopiq guruhi avtomatik yaratiladi

Keyingilar avtomatik a’zo

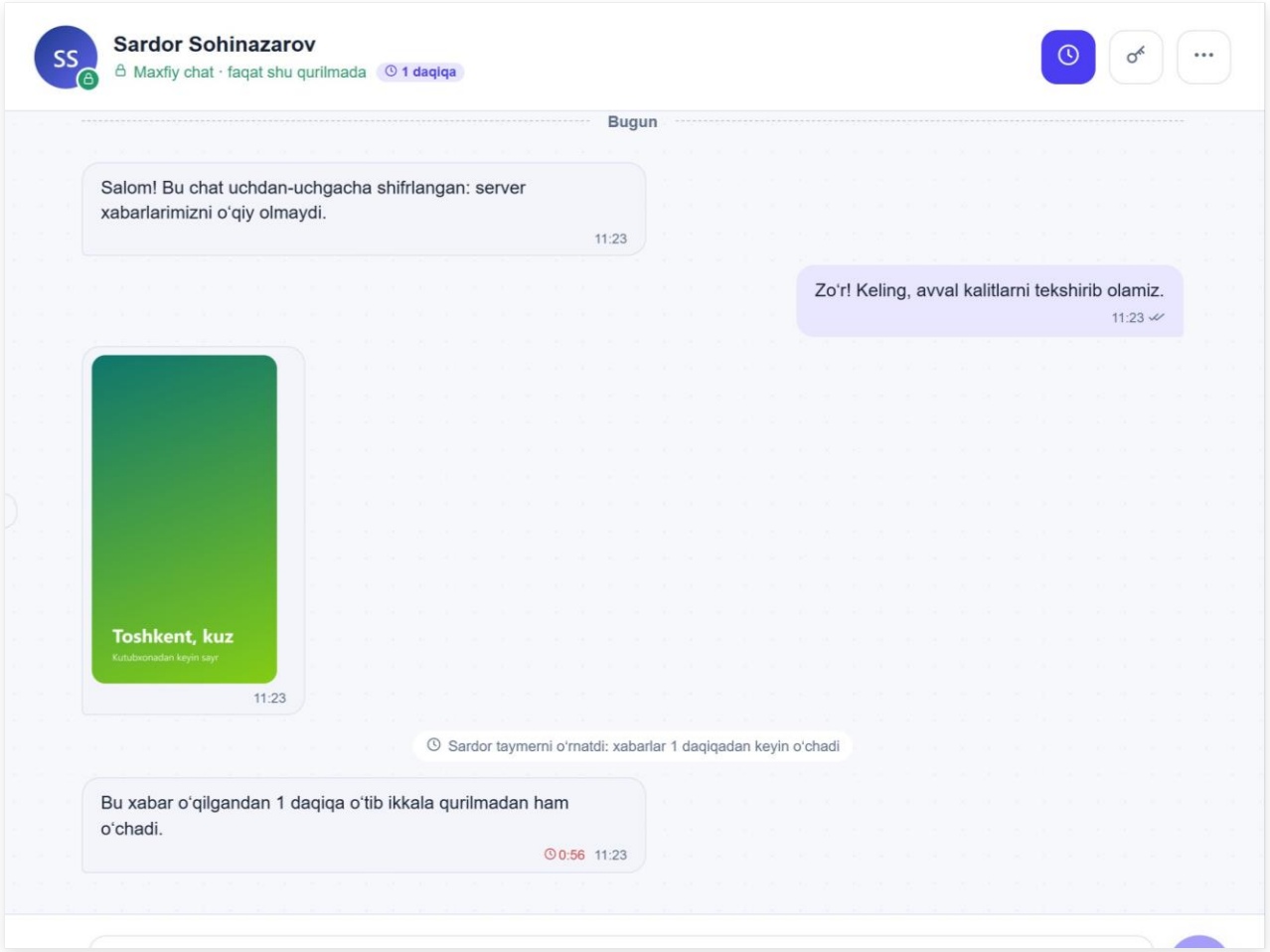
Ism yonida “tuit.uz ✓”, guruhga qo‘shiladi

Boshqalar — faqat taklif bilan

gmail.com va h.k.: admin yoki 128 bitli havola



Maxfiy chat: uchdan-uchgacha shifrlash



Protokol (Web Crypto API)

X25519 kalit kelishuvi — serverga faqat ochiq kalit

HKDF-SHA-256 ikki yo‘nalish uchun zanjir kalitlari

HMAC ratchet har xabarga yangi kalit, eskisi unutiladi

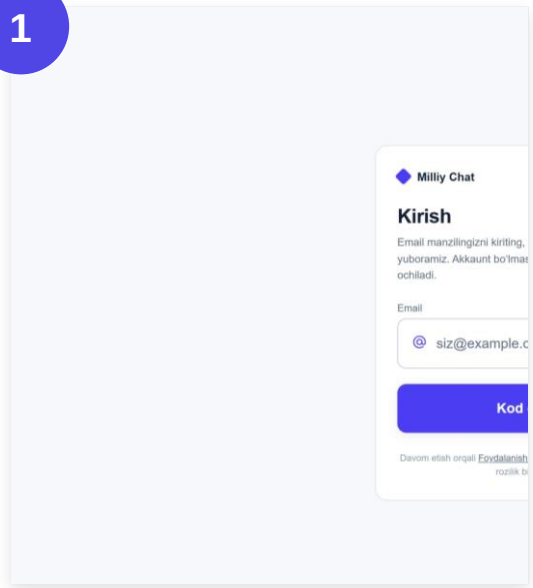
AES-256-GCM shifrlash; AAD = chat | yo‘nalish | seq

Serverdagi yagona iz (secret_messages):

```
Id | Seq | bytes | Ciphertext
 6 |   5 |   183 | AdBp3POJtrxqSydreRLgRu…
```

Dastur interfeysi: tizimga kirish

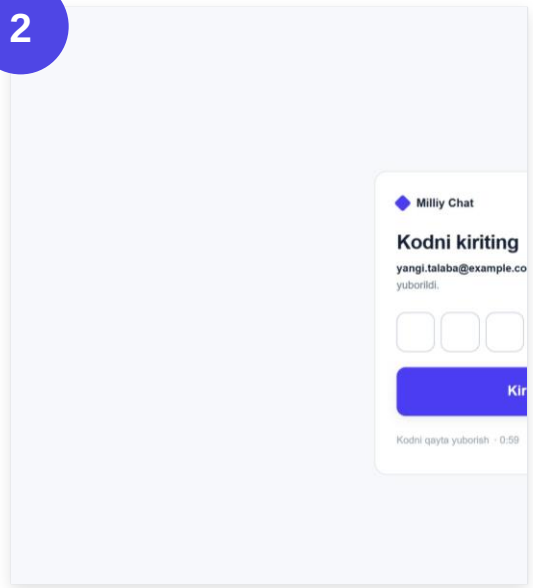
1



Kirish

E-pochta manzilini kiritish yoki Google orqali kirish

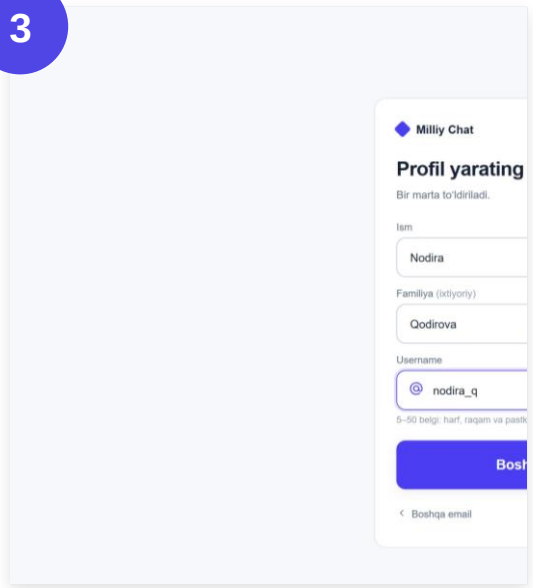
2



Tasdiqlash kodi

6 xonali bir martalik kod pochtaga yuboriladi

3

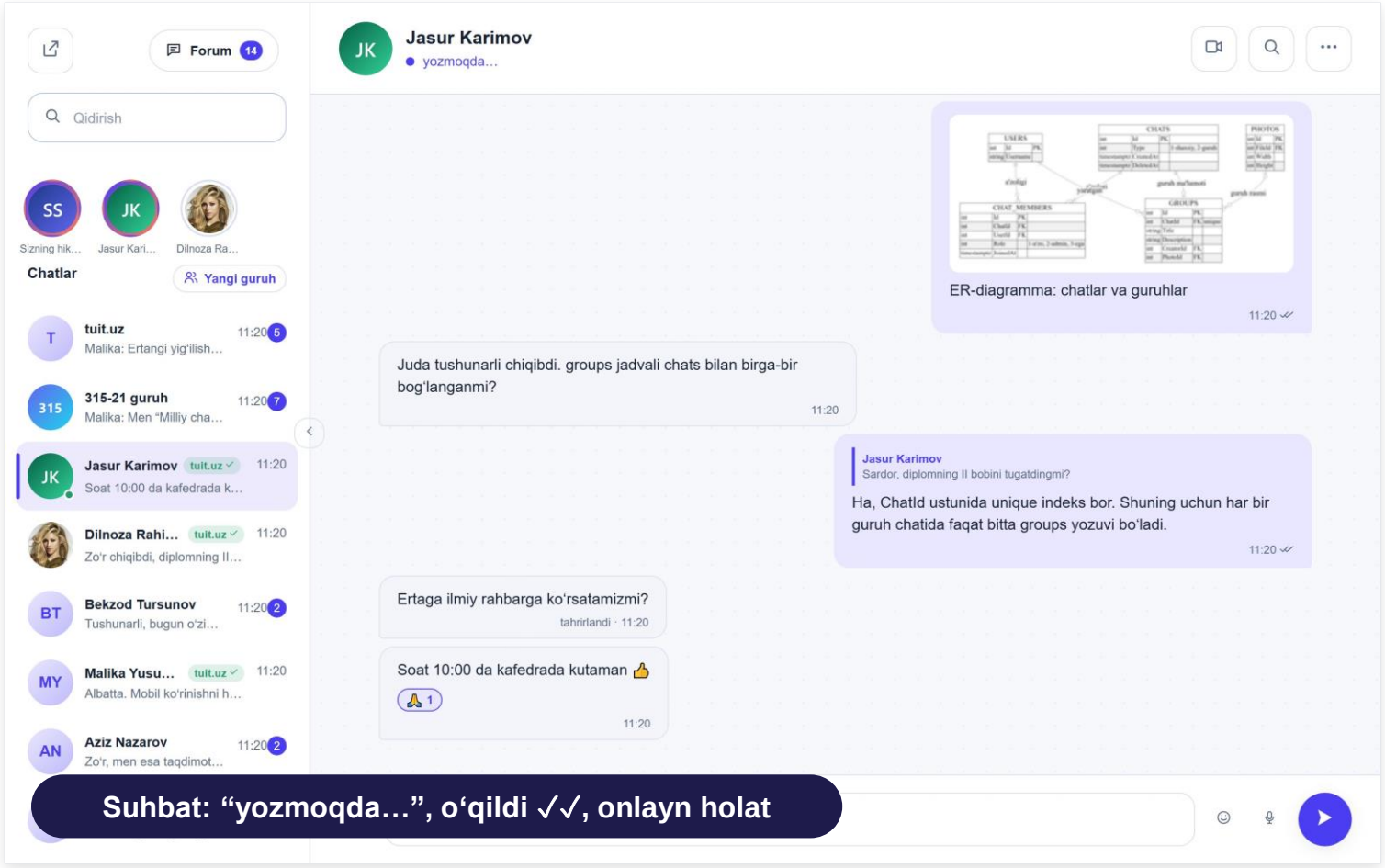


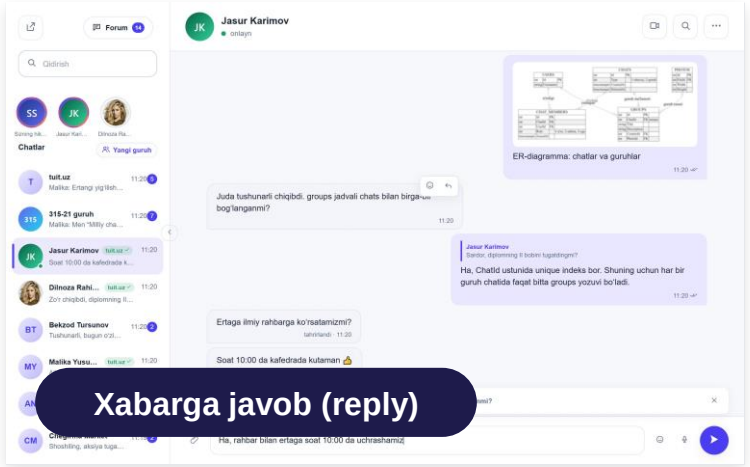
Ro‘yxatdan o‘tish

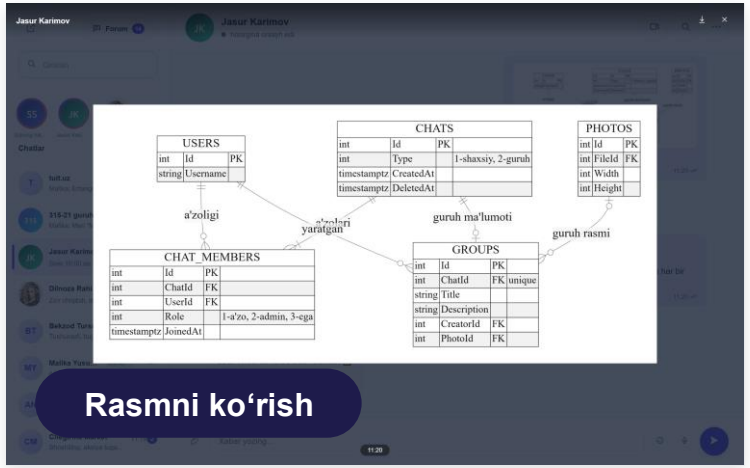
Yangi foydalanuvchi ismi va username’ini kiritadi

Parolsiz kirish: parol saqlanmaydi va sizib chiqmaydi — foydalanuvchi faqat pochtasiga egaligini tasdiqlaydi.

Dastur interfeysi: shaxsiy chat







Dastur interfeysi: guruh chatlari

Egasi

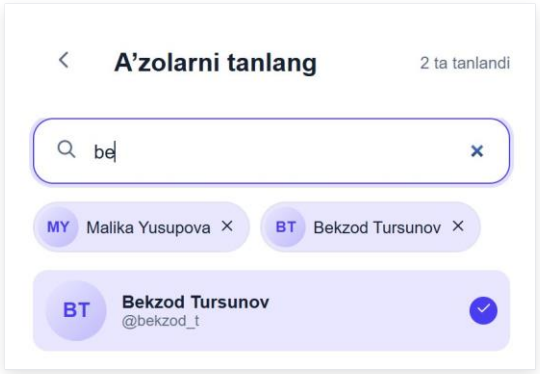
Adminlarni tayinlaydi, guruhni o‘chiradi

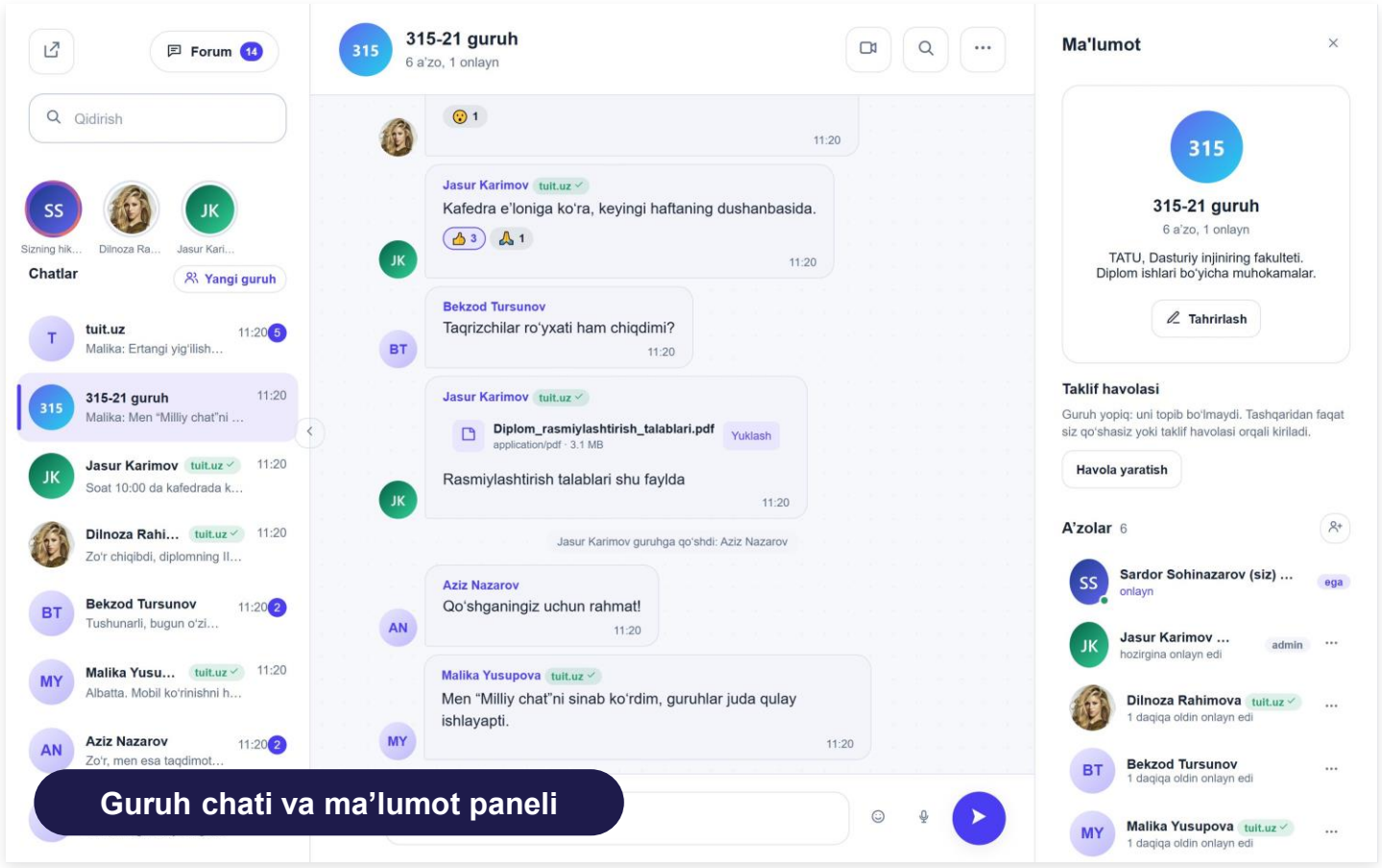
Administrator

A’zo qo‘shadi va chiqaradi, guruhni tahrirlaydi

A’zo

Xabar va fayllar yuboradi, guruhdan chiqadi

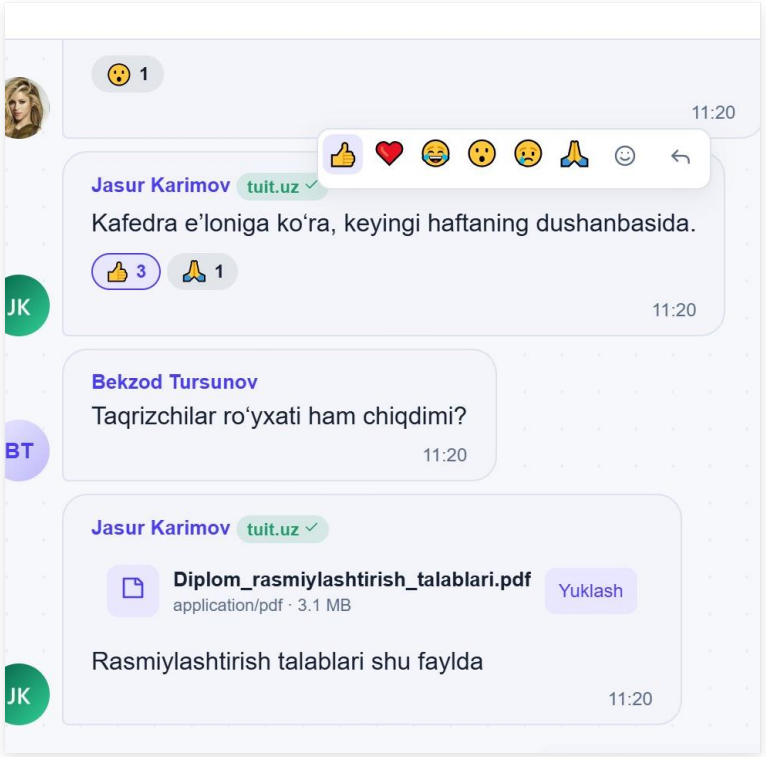




Reaksiyalar, lavhalar va bloklash

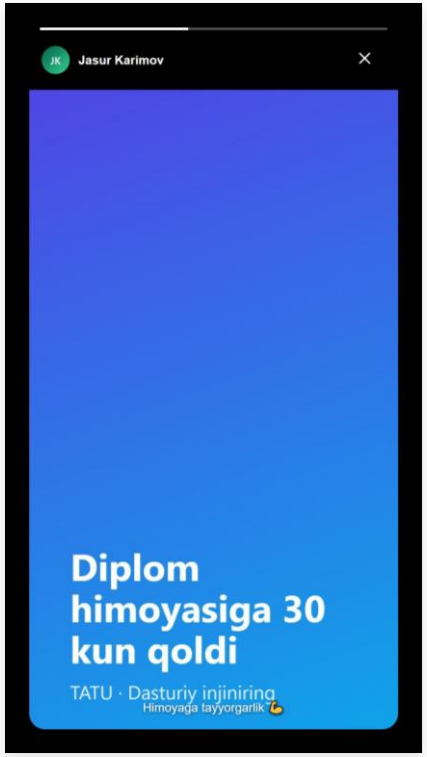
Reaksiyalar

Oltita emoji, real vaqtda barcha a’zolarda



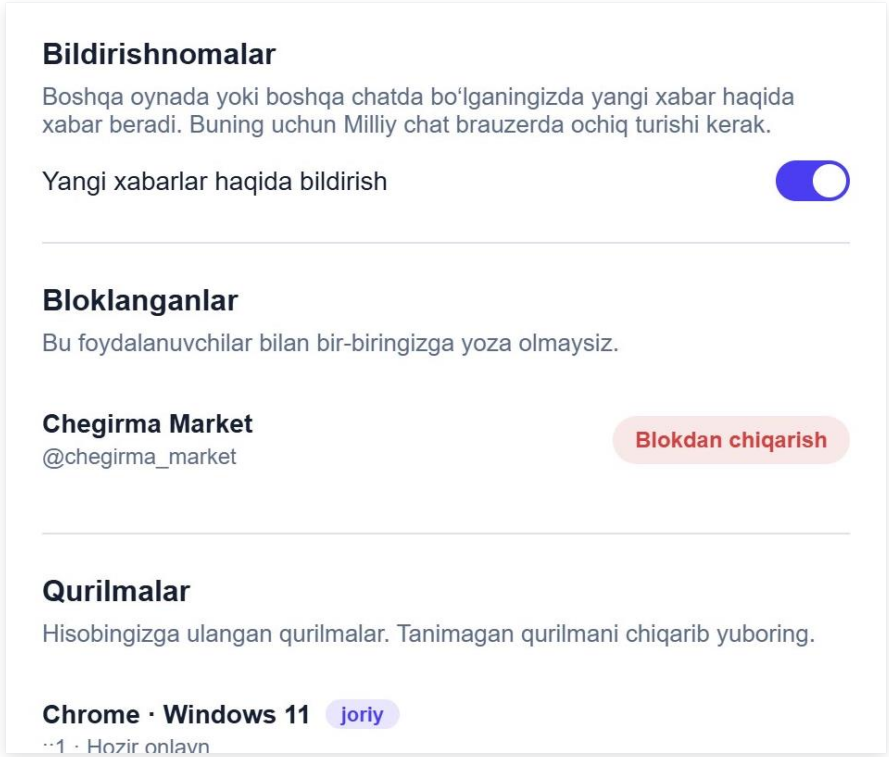
Lavhalar

24 soat, faqat obunalarniki

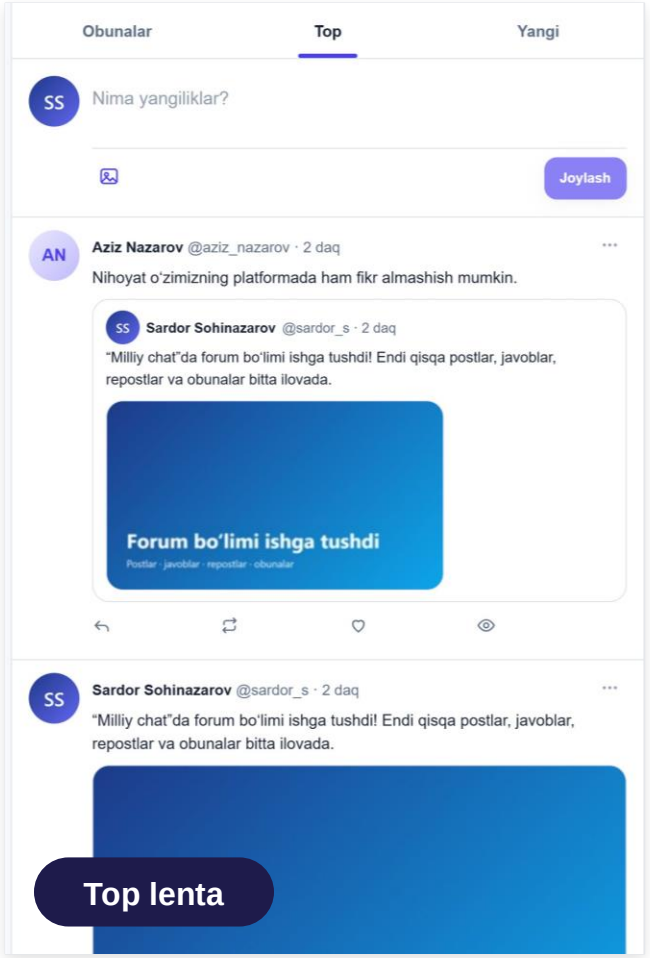


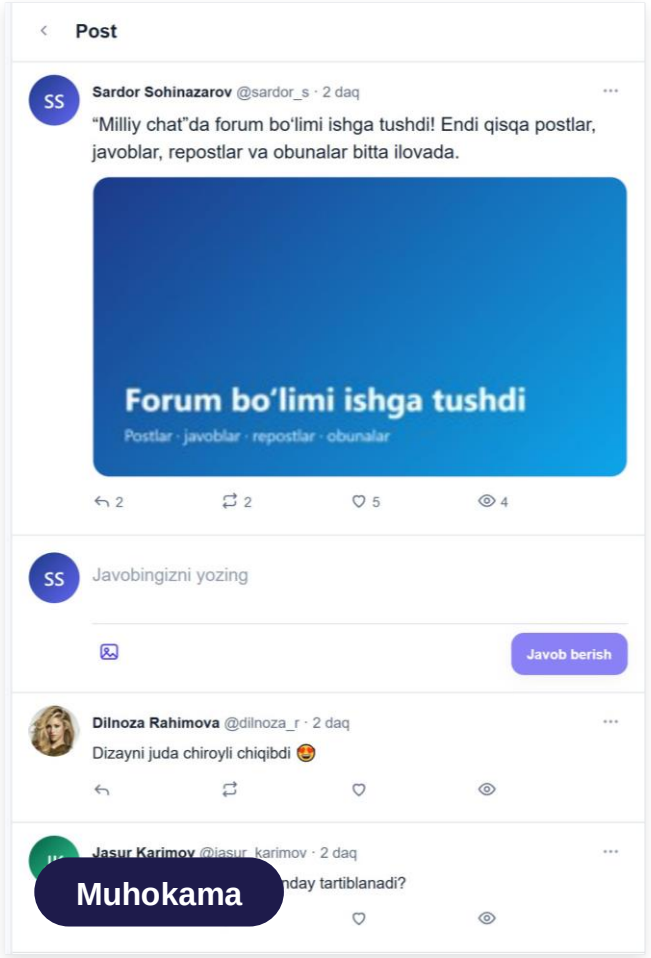
Bloklash

Ikki tomon ham yoza olmaydi; profilda ro‘yxati va bildirishnomalar



Forum: ochiq fikr almashish





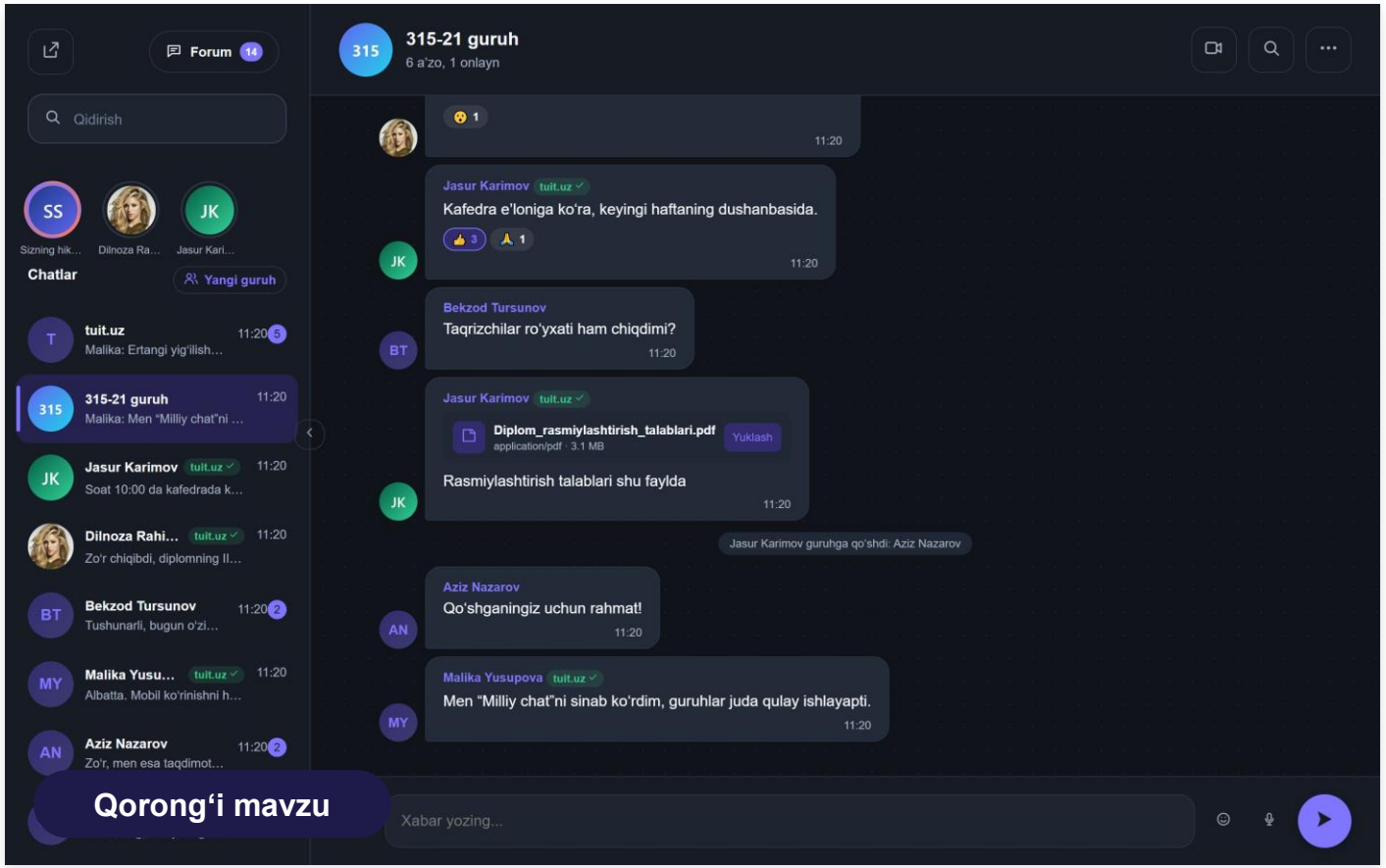
- 280 belgigacha post, 4 ta rasm yoki video
- Javob, repost, iqtibos, layk, obuna
- ForumHub: hisoblagichlar va bildirishnomalar real vaqtda
- Spamga qarshi: soatiga 10 ta muhokama, daqiqasiga 5 ta javob

Top lenta reytingi

$$S = (L + 2R + 3P + 0{,}01V) / (t + 2)^{1{,}5}$$

L — layk, R — javob, P — repost, V — ko‘rish, t — yoshi (soat)

Qorong‘i mavzu va mobil ko‘rinish







Bitta kod bazasi kompyuter, planshet va telefonda ishlaydi — o‘z dizayn tizimi va moslashuvchan maket.

Joylashtirish va testlash

104
REST API endpointi

720
unit test: 425 server + 295 klient

43
API integratsion tekshiruvi

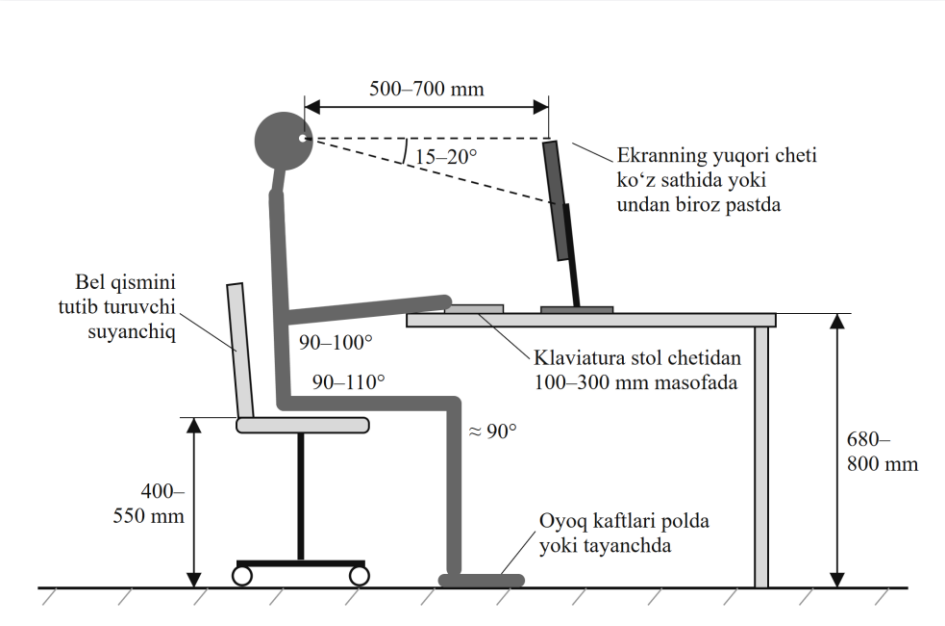
Docker
Render bulutida ishlaydi

- Konteynerlash texnologiyalaridan biri — Docker’dan foydalanildi
- GitHub’ga push qilinganda Render obrazni avtomatik qayta yig‘adi
- Maxfiy kalitlar kodda emas, muhit o‘zgaruvchilarida
- xUnit va Vitest: transliteratsiya, forum, maxfiy chat protokoli, tashkilotlar, reaksiya va bloklash
- Swagger UI orqali API hujjatlari va qo‘lda sinov

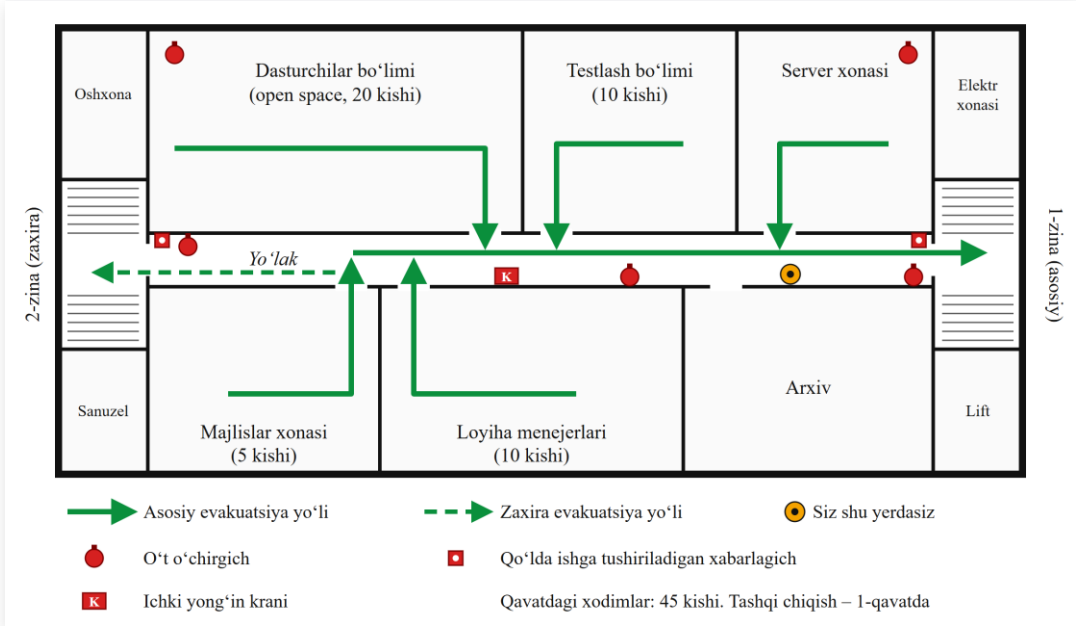
```
PS> dotnet test
Passed!  Failed: 0
Passed: 425, Total: 425

PS> ng test --watch=false
Test Files  20 passed
Tests      295 passed
```

Hayot faoliyati xavfsizligi



Ergonomik ish joyi va to‘g‘ri gavda holati



IT-ofis qavatining evakuatsiya reja-sxemasi

6 m² / 20 m³
bitta ish joyiga eng kam maydon va hajm

300–500 lk
stol yuzasining yoritilganligi

≥ 2 chiqish
har bir qavatda evakuatsiya uchun

OU-3 / OU-5
kompyuter xonalari uchun o‘t o‘chirgich

Xulosa va istiqbollar

Erishilgan natijalar

- Beshta messenjer, jumladan Humo tahlil qilindi, SignalR tanlandi
- 43 jadval, 104 API manzil, 720 ta test
- Chatlar, guruhlar, reaksiyalar, lavhalar, forum — real vaqtda
- Lotin↔kirill ko‘rinish va ikkala yozuvda qidiruv
- Tashkilot rejimi va maxfiy chatlar
- Elektron kooperatsiya portalining chat modulida qo‘llanildi

Istiqbollar

- Sun’iy intellekt yordamchisi
- Bozor: do‘konlar va savdo
- Kompaniyalar uchun auksion va tenderlar
- Blokcheyn: shaffof tender, aqlli shartnoma
- Qo‘ng‘iroqlar, kanallar va botlar
- OneID va mobil ilovalar

E’tiboringiz uchun rahmat!

Real vaqtda xabar almashish imkonini beruvchi “Milliy” xabarchi ishlab chiqish

S.O. Sohinazarov

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